Do Price Signals Predict Indian Stock Returns?

A Next-Day Forecast, a Monthly Strategy and a Survivorship-Free Test

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Abstract. In this study, I test whether signals built from past prices predict the returns of stocks listed on India's National Stock Exchange, first as a forecast of the next day's return and then as a monthly ranking of stocks. In the first model, a least squares model of each of 98 stocks' next-day returns, based on four moving-average features, is no better than benchmarks that use no model: in out-of-sample testing, the mean R^2 is −0.0155 and mean directional accuracy is 49.6%. The second model ranks stocks against each other each month on four price signals chosen on data up to 2018, and holds the top fifth. On today's Nifty 500 the strategy returns 32.0% a year from 2009 to 2026, but that universe holds only companies that survived the 17 years the model was backtested on. So I rebuilt it from the exchange's daily records of every stock that traded, including those that later went bankrupt, were delisted, were acquired or merged with another company, or shrank out of the top 500. On this universe, the same strategy returns 23.8% a year, against 16.8% for an equal-weighted portfolio of the same stocks and 16.0% for the Nifty 500. Survivorship bias added about 8 percentage points a year to both the strategy and its benchmark, and left the gap between them unchanged. In the 2019–2026 holdout, the strategy beats the benchmark by +8.3% a year ($t = 2.39$). The excess return is the momentum premium; it is about as large as that of NSE's own momentum index, and it comes mostly from stocks outside the largest 200.

*Code, data and the scripts that generate every number, table and figure in this paper are at https://github.com/arjunprashanth6129/price-prediction-linear-regression. An interactive summary of the results is at https://arjun-quant-projects.vercel.app/linear-regression.

1. Introduction

Technical analysis rests on the premise that the recent path of a price says something about its next move, and moving-average rules are its most common form. In one of the best-known versions, a short moving average that rises above a long one, a *golden cross*, is read as a signal to buy, and the reverse crossing, a *death cross*, as a signal to sell. Figure 1 shows both in Britannia Industries' share price in 2019: the 5-day average rose above the 20-day average on 28 August 2019 and fell back below it on 8 November 2019. Almost every charting platform draws these averages, and many retail investors rely on them for their trades.

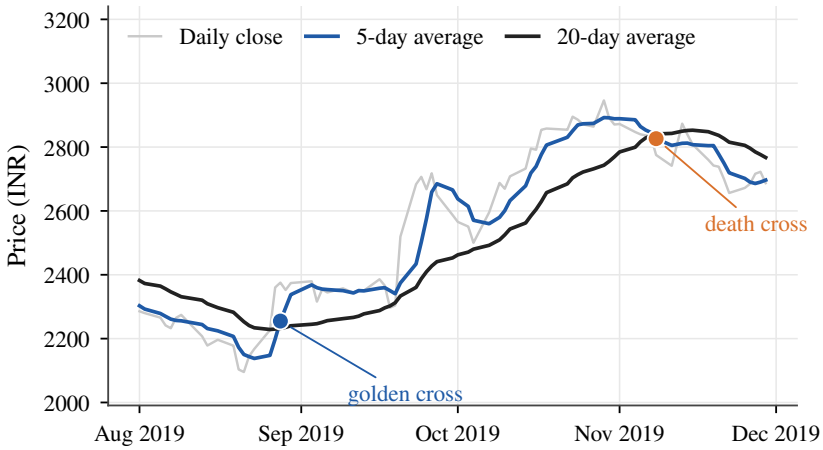


Figure 1. A golden cross and a death cross in Britannia Industries; the time frame is from August to November 2019. The golden cross is the day the 5-day moving average of the closing price rises above the 20-day average; the death cross is the day the 5-day moving average falls back below the 20-day. The ratio of the two averages, one of the four features used in the first part of this study, is above one between the two crossings and below one outside them.

The academic evidence on these rules is mixed and has weakened over time. Lo et al. (2000) put the visual patterns of technical analysis on a statistical footing and find that several carry incremental information about returns. Brock et al. (1992) found that moving-average and trading-range rules applied to the Dow Jones Industrial Average from 1897 to 1986 produced returns that standard null models of prices could not explain; Sullivan et al. (1999) re-examined the same universe of rules with a bootstrap that corrects for data snooping and found that the best rule's performance did not carry over into the following decade. In their survey of 95 modern studies, Park and Irwin (2007) report that most find profits until the early 1990s, but that many are affected by data snooping, ex post selection of rules and difficulties in measuring risk and transaction costs. Welch and Goyal (2008) show that most variables proposed as predictors of equity returns fail once they are evaluated on data not used to select them.

A separate section of the literature uses past prices to compare stocks with each other rather than to forecast one stock through time. Jegadeesh and Titman (1993) show that stocks with high returns over the previous three to twelve months go on to outperform those with low returns over the following months. George and Hwang (2004) find that a stock's closeness to its 52-week high explains much of that momentum effect, Han et al. (2016) build a trend factor from moving averages of several lengths, and Carhart (1997) adds momentum to the factors used to explain fund returns. Agarwalla et al. (2013) construct the same factors for Indian equities.

This paper asks a similar question. The first part of the study asks it in its narrow form: given a stock's recent price history, summarised by four moving-average features, does a linear model forecast the next day's return better than benchmarks that use no model? The second part asks it in the form the momentum literature

uses: ranked against each other once a month on price signals, do the highest-ranked stocks earn more than the rest, after the costs of trading them? Testing the second form exposed a problem with the data. The universe of stocks was today's list of large stocks, so every company in it had survived to today, and a backtest on survivors overstates returns (Brown et al., 1992). The final part of the study rebuilds the universe from the exchange's own records, which include the companies that later failed, and reruns the same strategy without changing it.

The first answer came back with negative returns. Across 98 stocks the mean out-of-sample R^2 is -0.0155 and mean directional accuracy is 49.6%, and no stock's accuracy is significantly above 50%. The second answer gave better results; on today's Nifty 500 the strategy returns 32.0% a year from March 2009 to June 2026. Survivorship accounts for about 8 percentage points of both the strategy's and the benchmark's return. On the corrected data the strategy returns 23.8% a year, against 16.8% for an equal-weighted portfolio of the same stocks and 16.0% for the Nifty 500, and in the holdout, which was run once after the model was altered, it beats that benchmark by +8.3% a year ($t = 2.39$). Its excess return is the momentum premium: once a momentum factor is controlled for, nothing is left.

The paper provides three insights. It provides out-of-sample evidence that daily moving-average features carry no useful information for a recent sample of Indian stocks. It tests a documented premium as a long-only strategy an investor could hold, taking into account Indian transaction costs and a one-day execution lag. And it also measures survivorship bias in an Indian backtest directly, by running one strategy on three universes: the bias inflated absolute returns by about 8 points a year, while the strategy's return relative to its benchmark stayed similar, with a difference of a few basis points.

I came to this question through an equity research internship at BugleRock Capital, where the contrast between fundamental analysis and technical trading came up repeatedly. Intrigued by how technical analysis worked, I felt that testing the most popular technical signal on data it had not seen was the most direct way to find out how much the evidence supports it.

Section 2 presents Part I. Section 3 sets out the design of the monthly strategy and its results on today's Nifty 500, and Section 4 rebuilds the universe and reruns it. Section 5 reports robustness checks and weak points, Section 6 discusses the findings and Section 7 their limitations. Section 8 concludes.

2. Part I: A next-day forecast

2.1 Data and design

The universe consists of 100 companies listed on the National Stock Exchange of India, chosen to span large, mid and small capitalisation. Daily closing prices from 1 June 2019 to 30 June 2026 were downloaded from Yahoo Finance. Two stocks returned no usable history and are excluded, leaving 98.

Let P_t denote a stock's adjusted close on day t. The four features use closes up to and including day t:

$$ma5_t = \frac{1}{5}\sum_{i=0}^{4} P_{t-i}, \qquad ma20_t = \frac{1}{20}\sum_{i=0}^{19} P_{t-i}, \qquad r_t = \frac{P_t}{P_{t-1}} - 1, \qquad ratio_t = \frac{ma5_t}{ma20_t}. \tag{1}$$

The ratio, when above one, depicts the golden-cross state and, when below one, the death-cross state, so it expresses the crossing rule as a continuous variable. For each stock i, the model

$$r_{i,t+1} = \beta_{i0} + \sum_{k=1}^{4} \beta_{ik}\, \tilde{x}_{ik,t} + \varepsilon_{i,t+1} \tag{2}$$

is estimated by ordinary least squares, where $\tilde{x}_{ik,t}$ is feature k standardised to zero mean and unit variance (Pedregosa et al., 2011). The first 80% of each stock's observations, June 2019 to January 2025 for a stock with a full history, form the estimation sample, and the remaining 20% the evaluation sample. The split is chronological, the standardisation uses estimation-sample statistics, and nothing is re-estimated during the evaluation sample.

Three statistics are computed on each stock's evaluation sample of n_i days. The out-of-sample R^2 compares the model with a constant forecast,

$$R^2_{OS,i} = 1 - \frac{\sum_t \left(r_{i,t+1} - \hat{r}_{i,t+1}\right)^2}{\sum_t \left(r_{i,t+1} - \bar{r}_i\right)^2}, \tag{3}$$

where $\bar{r}_i$ is the mean return over the evaluation sample, a stricter reference than the estimation-sample mean of Campbell and Thompson (2008). Directional accuracy is the share of days on which forecast and realised returns have the same sign, and the *edge* is the model's accuracy minus the accuracy of a rule that always predicts a rise, which equals the share b_i of up days in the evaluation sample. A forecast with no information that predicts a rise with probability one half has expected accuracy of 50%, so its expected edge is

$$\mathbb{E}\left[\text{Edge}_i \mid \text{uninformative forecast}\right] = \tfrac{1}{2} - b_i, \tag{4}$$

which is positive whenever up days are in the minority. Each stock's accuracy is tested against 50% with an exact binomial test, corrected for testing 98 hypotheses with the procedure of Benjamini and Hochberg (1995), and with the test of Pesaran and Timmermann (1992), which allows for the base rate of up days.

2.2 Results

Table 1 summarises the evaluation. The mean out-of-sample R^2 is −0.0155 and the median −0.0090. 81 of 98 R^2 values are negative and the remaining 17 positive R^2 values are all small (Figure 2, Panel A). A negative R^2 basically means that the fitted model forecasts next-day returns less accurately than a constant, which is the outcome expected when the regressors carry no information: the coefficients are fitted to noise in the estimation sample, and the noise does not recur.

Table 1
Part I: Out-of-sample performance across 98 stocks

Panel A. Fit of the return forecast	
Mean R^2_{OS}	−0.0155
Median R^2_{OS}	−0.0090
Range	−0.130 to +0.042
Stocks with $R^2_{OS} > 0$	17 of 98
Panel B. Direction of the forecast	
Mean directional accuracy	49.6%
Range of directional accuracy	41.3% to 55.3%
Mean share of up days (always-up benchmark)	47.8%
Mean edge over always-up (pp)	+1.85
Expected edge of an uninformative forecast (pp)	+2.22
Stocks with a positive edge	68 of 98
Panel C. Per-stock inference, 5% level	
Accuracy significantly above 50% (binomial)	0
Accuracy significantly below 50% (binomial)	7
Significant after Benjamini-Hochberg	0
Significant Pesaran-Timmermann statistic	5
Panel D. The moving-average ratio	
Stocks where `ma_ratio` has the largest coefficient	0 of 98
Positive coefficient	61 of 98
Mean absolute coefficient	0.0016

Statistics are computed on each stock's evaluation sample and averaged across stocks where stated. R^2_{OS} is defined in equation (3). The always-up benchmark's accuracy equals the share of up days. The expected edge of an uninformative forecast is $\frac{1}{2} - b_i$ from equation (4), averaged across stocks. Panel C counts stocks significant in two-sided tests at the 5% level; the Pesaran-Timmermann statistic is defined for 97 stocks. pp denotes percentage points.

Mean directional accuracy is 49.6%. The model beats the always-up rule for 68 of 98 stocks, by +1.85 percentage points on average, but this is not necessarily a positive result, because the benchmark is naive: it uses no model. Over the evaluation window, the median stock's price changed by −0.06%, so up days were in the minority for 73 stocks and the always-up rule averaged only 47.8%. By equation (4), a forecast with no information would have beaten it by +2.22 points on average. Figure 3 makes the comparison stock by stock: the model's edge tracks the edge an uninformative forecast would have earned, with a correlation of 0.55 across stocks.

In the binomial tests, no stock is significantly above 50% and 7 are significantly below, about the number expected by chance, and none of them survives the correction for multiple testing. The Pesaran-Timmermann statistic is significant at 5% for 5 of 97 stocks and for none after the same correction. The moving-average ratio, the feature that encodes the crossing rule, has the largest standardised coefficient for none of the 98 stocks.

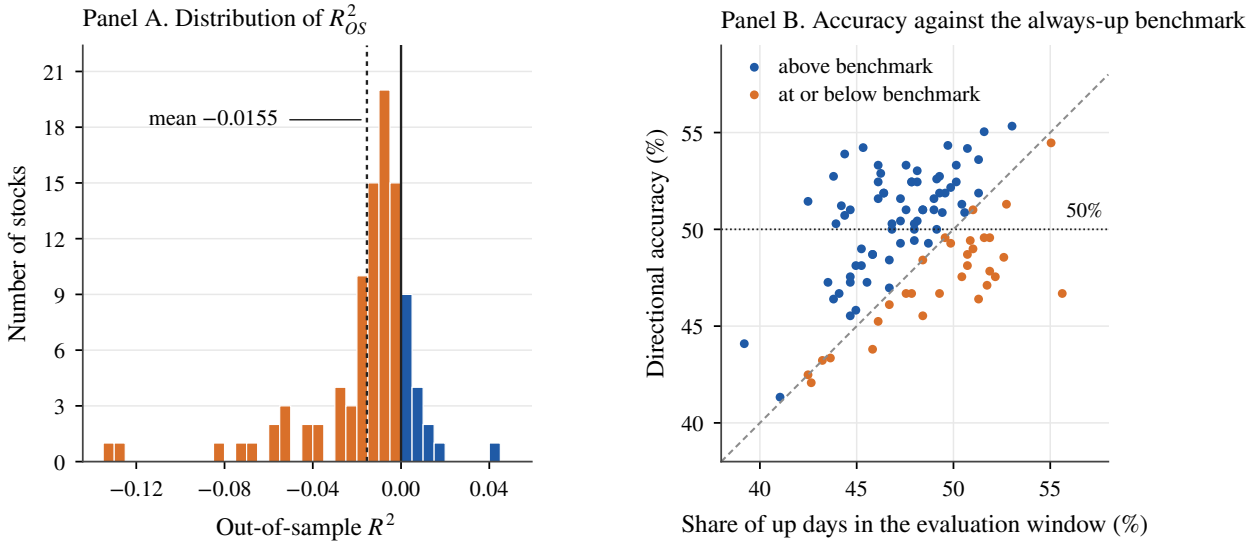


Figure 2. Part I: Explained variance and directional accuracy. Panel A shows the distribution of R^2_{OS} across stocks, with values above zero in blue; the solid line marks zero and the dashed line the mean. Panel B plots each stock's directional accuracy against the share of up days in its evaluation window, which represents the accuracy of the always-up rule. Points above the dashed 45-degree line beat that rule; the dotted line marks 50%.

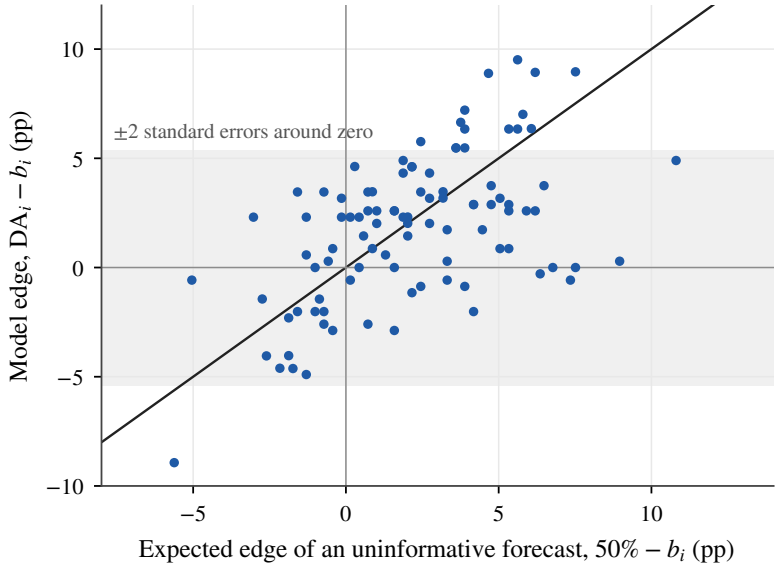


Figure 3. Part I: The model's edge against the edge of an uninformative forecast. Each point is a stock. The horizontal axis is $\frac{1}{2} - b_i$, the expected edge over the always-up rule of a forecast that predicts a rise with probability one half; the vertical axis is the model's realised edge. The solid line is the 45-degree line on which the two are equal. The shaded band is two standard errors of a single stock's accuracy around zero.

2.3 *Why the target is a return*

A natural alternative design forecasts the next day's price rather than its return. For example, for Reliance Industries, a model with the same features fitted to the next day's close attains an R^2 of 0.936 over the 347 evaluation days (Table 2), the kind of figure that suggests a strong model. The random walk, which forecasts tomorrow's close as today's, attains 0.963 without estimating anything, with a mean absolute error 27% lower. The reason is in the denominator of R^2. Over the evaluation sample the close had a standard deviation of INR 93.9, while a typical daily move was INR 18.1, so any forecast close to the previous close explains most of the variance of the level, whether or not it anticipates the change.

Table 2
Part I, Reliance Industries: Level and return forecasts over the evaluation sample

	Price model	Random walk	Return model
MAE	INR 18.23	INR 13.26	0.0097
RMSE	INR 23.70	INR 18.08	0.0133
R^2 against the test mean	0.9363	0.9629	−0.0080
Directional accuracy (%)	47.0	51.6	51.0

The price model and the return model use the same four features, the same split and the same estimator; only the target differs. The random walk forecasts tomorrow's close as today's close; because it never forecasts a rise, its directional accuracy is the share of days on which the price did not rise. MAE and RMSE for the first two columns are in rupees; the return model's errors are fractions of the price.

The last row of Table 2 scores direction, and there the high R^2 counts for nothing: the price model calls the next day's move correctly on 47.0% of days, worse than the probability of heads on a coin flip. Figure 4 plots each forecast against the outcome. The level forecasts lie along the diagonal because prices change little from one day to the next, while the return forecasts form a narrow vertical band within 0.47% of zero. For an autocorrelated series such as a price level, the right reference for R^2 is the random walk, not the sample mean.

2.4 *What the null result suggested*

The results are consistent with the weak form of the efficient market hypothesis (Fama, 1970) at the daily horizon, but the design limited what they could show. Each stock was forecast on its own history, one day ahead, with no portfolio and no costs, and about 345 evaluation days per stock gave too little power to detect an effect of one or two percentage points of accuracy. The literature on price signals that do carry information measures them differently: across stocks rather than through time, over months rather than days, and as the return of a portfolio rather than the accuracy of a forecast (Jegadeesh and Titman, 1993). Part II of this study tests price signals in that form.

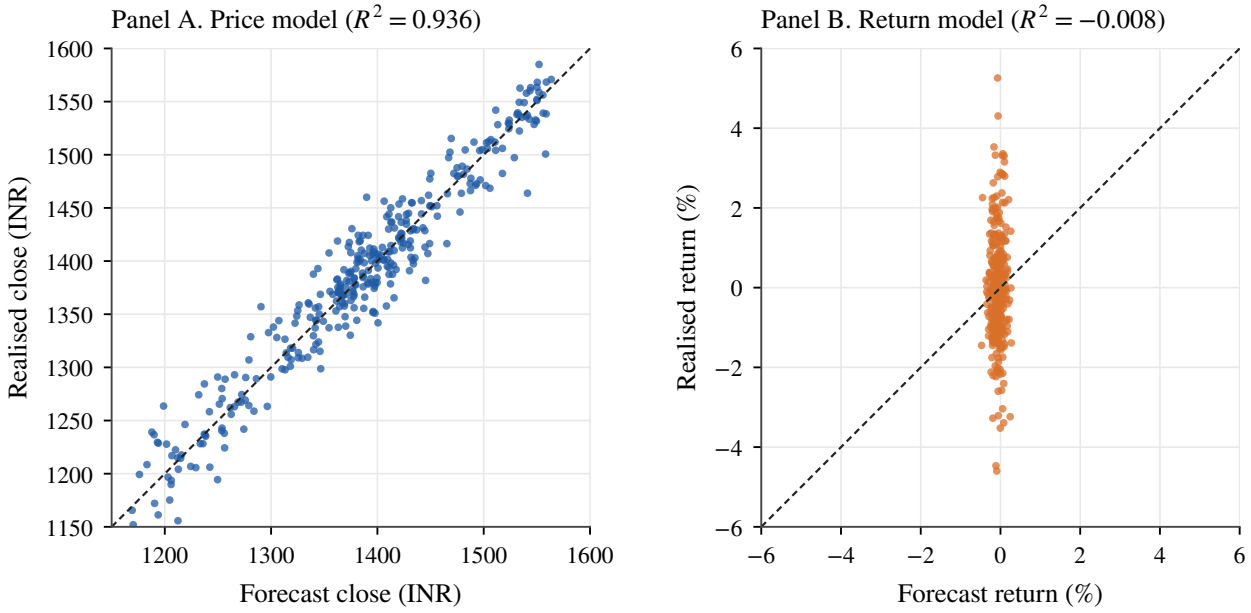


Figure 4. Part I: Forecast against realised values for Reliance Industries over the same 347 evaluation days. Panel A is the price model and Panel B the return model; the dashed line marks perfect forecasts.

3. Part II: A monthly strategy

3.1 Universe and data

The universe is the Nifty 500, as it stood when the data were downloaded in October 2026, 499 stocks with a price history, with daily closes and volumes from January 2005 to June 2026 from Yahoo Finance. The adjusted closes include dividends.[1] I cleaned the prices before any signal was computed: one-day spikes that reverse within five sessions were removed as bad ticks, a stock's history restarts after a suspension of 60 sessions or more, and splits, bonuses and demergers that Yahoo leaves in its adjusted series were repaired.

At each month-end a stock is eligible if it has 252 sessions of history, traded on at least 90% of the last 252 sessions, and is not among the least liquid 20% of the remaining stocks by six-month median traded value. Each check uses data up to that month-end only.

3.2 Signals and how they were chosen

I fixed a list of 10 candidate signals (Table 3). They are price-based signals with a published rationale: 12-1 momentum (Jegadeesh and Titman, 1993), closeness to the 52-week high (George and Hwang, 2004), the ratio of a moving average to the price at five horizons, which generalises Part I's feature and follows Han et al. (2016), low volatility, one-month reversal and a volume shock. On each month-end every signal is winsorised at the 1st and 99th percentiles across eligible stocks and standardised to a z-score.

Table 3
The candidate signals and the selection rule

Signal	Definition	Mean IC	t	Selected
12-1 momentum	$P_{t-21}/P_{t-252} - 1$	+0.044	+2.55	yes, +
Close / 52-week high	$P_t/\max_{252} P$	+0.044	+2.39	yes, +
5-day trend	MA_5/P_t	+0.022	+1.90	no
20-day trend	MA_{20}/P_t	+0.006	+0.56	no
50-day trend	MA_{50}/P_t	−0.007	−0.52	no
100-day trend	MA_{100}/P_t	−0.027	−1.79	no
200-day trend	MA_{200}/P_t	−0.037	−2.18	yes, −
Low volatility	$-\sigma_{252}$	+0.026	+1.49	no
1-month reversal	$-(P_t/P_{t-21} - 1)$	0.000	−0.02	no
Volume shock	$\log(\overline{V}_{20}/\overline{V}_{252})$	+0.021	+2.01	yes, +

P_t is the close at month-end t, with lags in trading sessions; MA_L is the L-session moving average, σ_{252} the volatility of daily returns over 252 sessions and $\overline{V}_n$ the mean daily traded value over n sessions. The information coefficient (IC) is the rank correlation across eligible stocks between the signal and the next month's return; t is the Newey-West t-statistic of its mean with 6 lags, on signal dates from January 2006 to October 2018. A signal is selected if $|t| \geq 2$, with the sign of its mean IC.

The selection rule was fixed before the data were examined: a signal enters the strategy if the mean of its monthly rank information coefficient (IC) on the development period has a Newey-West t-statistic (Newey and West, 1987) of at least 2 in absolute value, with the sign of that mean. Four signals passed: 12-1 momentum ($t = +2.55$), closeness to the 52-week high ($t = +2.39$), the volume shock ($t = +2.01$) and the 200-day trend ($t = -2.18$), which enters with a negative sign, so a close above its 200-day average scores well (Figure 5).

[1] Part I's paper stated that returns excluded dividends. Comparing Yahoo's adjusted and unadjusted closes around ITC's ex-dividend dates shows that the adjusted series includes them.

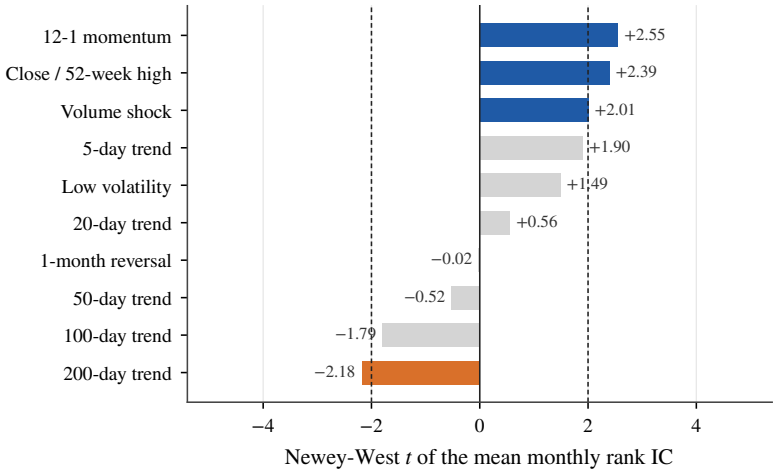


Figure 5. Signal selection on the development period. Each bar is the Newey-West t-statistic of a signal's mean monthly rank IC. Signals beyond the dashed lines at ±2 are selected, in blue with a positive sign and orange with a negative one. This is the selection that fixed the strategy.

3.3 Portfolio, costs and timing

The score of a stock is the average of its four selected z-scores, each with the sign of its IC. At each month-end the strategy buys the top fifth of eligible stocks by score in equal weights and holds them, with weights drifting with prices, until the next month-end. It is long only, since an investor cannot hold a short position overnight in the Indian cash market. The signal uses the month-end close and the trade is made at the close of the next session, so the price that produced a signal is never the price paid.

Each trade is charged the delivery costs in force on the NSE since October 2024: securities transaction tax of 0.1% on both purchases and sales, stamp duty of 0.015% on purchases, exchange and SEBI fees with goods and services tax on them, no brokerage, and a bid-ask spread of 5, 15 or 35 basis points for the most, middle and least liquid third of the universe. The benchmark is an equal-weighted portfolio of every eligible stock (EW), rebalanced on the same dates and charged the same costs. It shares the strategy's universe, so any bias in the universe affects both, and the comparison between them is the test of the signals.

3.4 Protocol

The development period runs from March 2009 to December 2018 and the holdout from January 2019 to June 2026. Every choice described above, including the cost schedule and the list of variants to report, was committed to a GitHub repository before any backtest was run, and while the specification was being fixed the code that loads prices stopped at December 2018. One choice changed on development data. At first, a walk-forward Fama and MacBeth (1973) regression of returns on the four signals was the primary model; on the development period its forecasts had a negative rank IC, because slopes estimated through the 2008–09 crash put large offsetting weights on correlated signals. I replaced it with the equal-weighted score before the holdout was opened. The holdout was then run once again.

3.5 Inference

Returns are monthly and net of costs. The return over the equal-weighted baseline strategy (EW) is the mean monthly difference between the strategy and EW, annualised, and its t-statistic uses Newey-West standard errors with 6 lags. Alpha against EW is the annualised intercept of a regression of the strategy's excess return

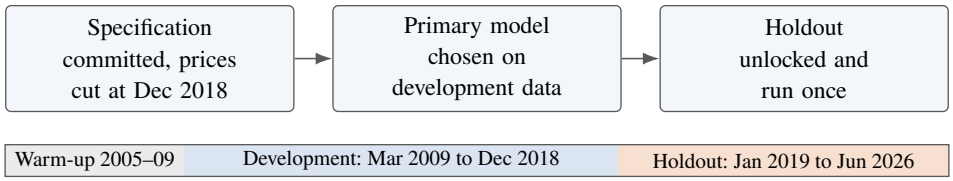


Figure 6. The order of the work. The top row is the sequence of commits in the repository; the bar below is the sample. Signals are selected on signal dates from January 2006 to October 2018, whose next-month returns end within the development period. Warm-up years provide the 252 sessions of history the signals need.

on EW's. Because 20 configurations were evaluated, including the variants in Section 5, the Sharpe ratio is also reported as the deflated Sharpe ratio of Bailey and López de Prado (2014), the probability that it exceeds the best Sharpe ratio expected from that many trials with no skill, and with a 95% interval from 5,000 draws of a stationary bootstrap (Politis and Romano, 1994) with a mean block of 6 months.

The strategy is compared with 1,000 random portfolios of the same size drawn from the same eligible stocks each month, and its long-short spread with 500 spreads built from scores shuffled across stocks, both before costs. The excess returns are regressed on the market, size, value and momentum factors of Agarwalla et al. (2013). As a test for look-ahead, the same engine ranks stocks on their realised next-month return, which must produce an implausible Sharpe ratio, and on scores one month stale, which must do somewhat worse than live scores. As a test that the signals are built correctly, the 12-1 momentum spread is compared with the published momentum factor. Unit tests check the engine, including one that changes every price after a date and confirms that no signal, weight or score dated before it changes.

3.6 Results on today's Nifty 500

Table 4 reports the results on today's Nifty 500. The strategy returns 32.0% a year over the full sample with a Sharpe ratio of 1.15, against 25.0% for EW and 14.8% for the Nifty 500 price index. In the holdout it beats EW by +8.5% a year ($t = 2.67$). These numbers are too high to be believed as they stand. EW, a portfolio with no signal at all, beat the index by about ten percentage points a year, and an equal-weighted portfolio should not outperform a cap-weighted index by that much for seventeen years. Section 4 shows the reason behind it.

Table 4
The strategy on today's Nifty 500, net of costs

	Development 2009–2018	Holdout 2019–2026	Full 2009–2026
Strategy CAGR	31.0	33.3	32.0
Equal-weight universe (EW) CAGR	26.9	22.7	25.0
Nifty 500 price index CAGR	16.1	13.0	14.8
Strategy Sharpe ratio	1.14	1.16	1.15
EW Sharpe ratio	0.81	0.80	0.81
Strategy maximum drawdown	−27.9	−36.9	−36.9
Return over EW, a year (t)	+2.4 (0.47)	+8.5 (2.67)	+5.0 (1.60)
Alpha against EW, a year (t)	+8.5 (2.73)	+9.9 (3.19)	+9.2 (4.09)
Deflated Sharpe ratio	0.99	0.95	1.00
Cost drag, a year	2.6	2.5	2.5

CAGR is the compound annual growth rate. Sharpe ratios use the risk-free rate of Agarwalla et al. (2013). EW is an equal-weighted portfolio of every eligible stock, rebalanced monthly and charged the same costs. The index is the Nifty 500 price index, the only series Yahoo provides. Return over EW is the annualised mean monthly difference and alpha the annualised intercept of a regression on EW, both with Newey-West t-statistics. The deflated Sharpe ratio allows for 20 configurations evaluated.

4. Part III: A survivorship-free test

4.1 The problem

A company enters today's Nifty 500 by being among the largest listed companies now. Companies that were large in 2010 and have since collapsed, been suspended or been delisted are not in it, so a portfolio drawn from today's list in 2010 holds only the stocks that went on to do well enough to remain listed and large (Brown et al., 1992). For an equal-weighted benchmark the effect is simple: its return is inflated. For a momentum strategy it could be worse, because momentum buys past winners, and the past winners that later failed are exactly the stocks the list leaves out.

NSE Indices publishes a Nifty500 Equal Weight index whose history it computed from each date's actual constituents, which makes the size of the problem measurable. From March 2009 to June 2026 the equal-weighted portfolio of today's Nifty 500 beat that index by +6.4% a year, net of costs (Figure 7).

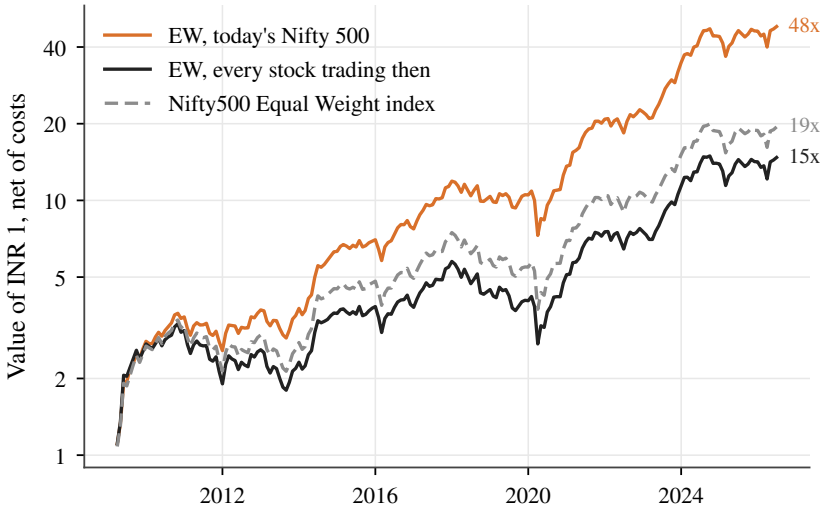


Figure 7. Survivorship bias in the equal-weighted benchmark. The value of one rupee, net of costs, in the equal-weighted portfolio of today's Nifty 500, in the equal-weighted portfolio of every stock trading at the time, and in NSE's Nifty500 Equal Weight index, which is built from each date's actual constituents.

4.2 Rebuilding the universe from exchange records

NSE publishes a file for each trading session listing every security that traded, with its close, volume and traded value. I downloaded all 5,304 sessions from January 2005 to June 2026, kept the equity series, excluded exchange-traded funds, and mapped renamed symbols to their current names, using NSE's list of symbol changes and, from 2011, matching ISINs. The panel keeps 2,581 stocks: every equity that was among the 1,000 most traded at some month-end. The universe is rebuilt at each month-end from what was trading then: the 500 most traded stocks by six-month median traded value, which is about the size of the Nifty 500, followed by the same eligibility rules as before. That leaves 400 stocks each month, and 1,359 different stocks pass through the universe over the sample. Traded value ranks stocks by investability rather than by market capitalisation, because a history of shares outstanding is not freely available, and it needs no adjustment for splits.

The exchange's prices are transaction prices, with no adjustment for corporate actions. For the period from January 2010 onward, corporate action information was obtained from NSE's corporate action records. Listing every unexplained one-day fall of more than 40% in a liquid stock found 6 splits and bonuses missing

from NSE's records, each of which was verified against the company's filing. Before 2010 the records did not exist, so capital changes are found from prices: a fall close to a standard split or bonus ratio in a liquid stock, or a fall of more than 80% in any stock. Both rules were checked on 2010–2026, where NSE's records give the answer: the first flags 670 moves of which 664 were recorded corporate actions, the second 256 of which 252 were. Large falls that fit neither rule are kept as real returns.

A stock that stops trading needs a final return. A merger or a voluntary delisting pays its holders the acquirer's shares or an exit offer, so its final return is set to zero. A compulsory delisting, a liquidation, or an exit after the stock had been moved to NSE's trade-for-trade segment or lost half its price in its last year is treated as distress and given a final return of −30%, the convention of Shumway (1997), with 0% and −100% reported as bounds. A suspension of 60 sessions or more that ends in resumed trading books the actual price change across the gap.

Two comparisons assess the rebuilt return series. For stock-months covered by both sources from 2010 onward, the rebuilt monthly return is within one percentage point of Yahoo's in 99.1% of the 70,368 observations. Before 2010, the share is 92.1%; this comparison is less direct because NSE's records do not include dividends for those years. The rebuilt universe's equal-weighted portfolio earns 17.5% a year before costs, compared with 18.7% for NSE's Nifty500 Equal Weight index. Their monthly returns have a correlation of 0.994, indicating that the portfolio generally moves with the index. Its lower return is consistent with the conservative treatment of stock exits and the omission of dividends where records are unavailable.

4.3 Protocol

The rules in this section were fixed by checking the data alone, against Yahoo, against NSE's index and against NSE's own records, and were fixed before any strategy return on the new data was computed. The strategy was then run once, unchanged. Because the holdout had already been seen in Part II, this is a replication of the same strategy on corrected data, not a new out-of-sample test.

4.4 Results

Table 5 and Figure 8 report the results. Over the full sample the strategy returns 23.8% a year with a Sharpe ratio of 0.80, against 16.8% and 0.48 for EW and 16.0% and 0.55 for the Nifty 500 total return index. Its worst drawdown, −37.0%, is shallower than EW's −58.1%. Its return over EW is +5.2% a year (t = 1.43) and its alpha against EW +8.4% (t = 3.46). In the holdout the return over EW is +8.3% (t = 2.39) and the alpha +9.6% (t = 2.64). The development period is weaker on its own: the return over EW of +2.8% has a t-statistic of only 0.48, and only the alpha (t = 2.12) is significant there. The deflated Sharpe ratio is 0.95 for the full sample and 0.89 for the holdout. The strategy holds about 80 stocks, replaces about 417% of the portfolio a year and pays 2.6% a year in costs; its return over EW falls to zero only at 3.6 times the cost schedule.

4.5 How much was survivorship

Table 6 shows the results from running the same strategy on three universes. Moving from Yahoo's prices to NSE's, with today's Nifty 500 names in both, the strategy's return changes only by 0.3 percentage points a year, so the data source is not what affects the strategy. Moving from today's names to every stock that was trading at the time lowers the strategy's return by 8.1 points a year and EW's by 8.3 points. The bias was large, and it lifted the strategy and its benchmark by almost exactly the same amount: the return over EW is +5.0%, +5.1% and +5.2% a year on the three universes. The concern that momentum would profit most from a universe of survivors does not show up in these data. The likely reason appears in Section 5: companies

Table 5
The strategy on the survivorship-free universe, net of costs

	Development 2009–2018	Holdout 2019–2026	Full 2009–2026
Strategy CAGR	21.3	27.3	23.8
Equal-weight universe (EW) CAGR	16.4	17.3	16.8
Nifty 500 total return CAGR	17.4	14.2	16.0
Nifty500 Momentum 50 CAGR	23.6	19.7	21.9
Strategy Sharpe ratio	0.69	0.93	0.80
EW Sharpe ratio	0.43	0.58	0.48
Strategy maximum drawdown	−34.7	−37.0	−37.0
Return over EW, a year (t)	+2.8 (0.48)	+8.3 (2.39)	+5.2 (1.43)
Alpha against EW, a year (t)	+6.9 (2.12)	+9.6 (2.64)	+8.4 (3.46)
Deflated Sharpe ratio	0.85	0.89	0.95
Cost drag, a year	2.7	2.6	2.6

Every NSE stock trading at each month-end, rebuilt as in Section 4.2. The Nifty 500 and Nifty500 Momentum 50 are NSE Indices' total return series, which pay no trading costs. Other definitions are as in Table 4.

on their way to failure leave the 500 most traded stocks well before they stop trading, so the strategy rarely holds them when they do.

Table 6
One strategy on three universes, March 2009 to June 2026

	CAGR		Return over	Four-factor alpha	EW minus
Universe	Strategy	EW	EW (t)	over EW (t)	Nifty500 EW
Today's Nifty 500, Yahoo prices	32.0	25.0	+5.0 (1.60)	+0.2 (0.11)	+6.4
Today's Nifty 500, NSE prices	31.7	24.7	+5.1 (1.62)	+0.1 (0.04)	+6.1
Every stock trading at the time, NSE	23.8	16.8	+5.2 (1.43)	+0.2 (0.09)	−1.9

All returns are net of costs and in percent a year. The four-factor alpha is the annualised intercept of a regression of the strategy's monthly return minus EW's on the factors of Agarwalla et al. (2013), to December 2025. The last column is EW's compound annual return minus that of NSE's Nifty500 Equal Weight total return index, which NSE computes from each date's actual constituents.

4.6 Skill, luck or leakage

Table 7 reports the checks of Section 3 on the survivorship-free data. The score's mean monthly rank IC is +0.055 over the full sample (t = 3.85) and +0.042 in the holdout (t = 2.13). Before costs, the strategy's Sharpe ratio of 0.92 exceeds that of all 1,000 random portfolios, whose 95th percentile is 0.56, and its long-short spread beats 100.0% of the shuffled spreads. Ranking on the realised return produces a Sharpe ratio of 5.2, as it should, and scores one month stale do slightly worse than live ones, 0.74 against 0.80, so information reaches the portfolio at the time it should and not before. The 12-1 momentum spread built here has a correlation of 0.861 with the momentum factor of Agarwalla et al. (2013). Returns rise steadily across the five quintiles of the score (Figure 9).

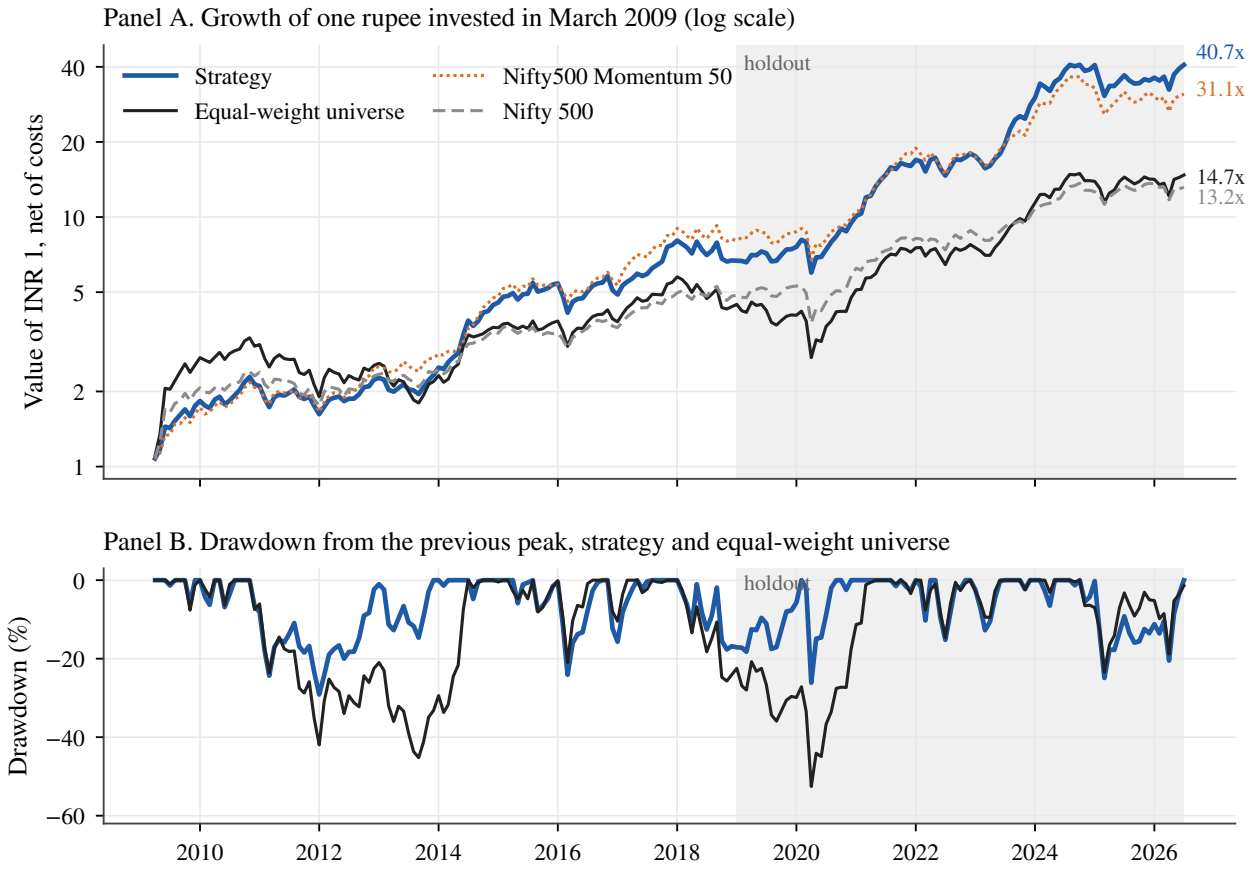


Figure 8. The strategy on the survivorship-free universe, net of costs. Panel A shows the value of one rupee invested in March 2009, on a logarithmic scale, with the final multiple at the right; the shaded area is the holdout. The Nifty 500 and the Nifty500 Momentum 50 are NSE Indices' total return series, which pay no costs. Panel B shows the fall of the strategy and EW from their previous peaks.

Table 7
Skill, luck and leakage on the survivorship-free universe

	Development	Holdout	Full
Mean monthly rank IC of the score (t)	+0.068 (3.46)	+0.042 (2.13)	+0.055 (3.85)
Random portfolios beaten, of 1,000	100%	100%	100%
Strategy Sharpe ratio before costs	0.81	1.04	0.92
95th percentile of random Sharpe ratios	0.51	0.69	0.56
Placebo spreads beaten, of 500	99.2	99.6	100.0
Sharpe ratio ranking on the realised return	4.9	5.9	5.2
Sharpe ratio with scores one month stale	0.64	0.85	0.74

The rank IC is the monthly Spearman correlation across eligible stocks between the score and the next month's return, with a Newey-West t-statistic. Random portfolios hold the same number of stocks, drawn at random from the eligible stocks each month; placebo spreads use the strategy's scores shuffled across stocks. Both are before costs. The last two rows are the leak checks: the same engine ranking stocks on their realised next-month return, and on scores one month old, net of costs.

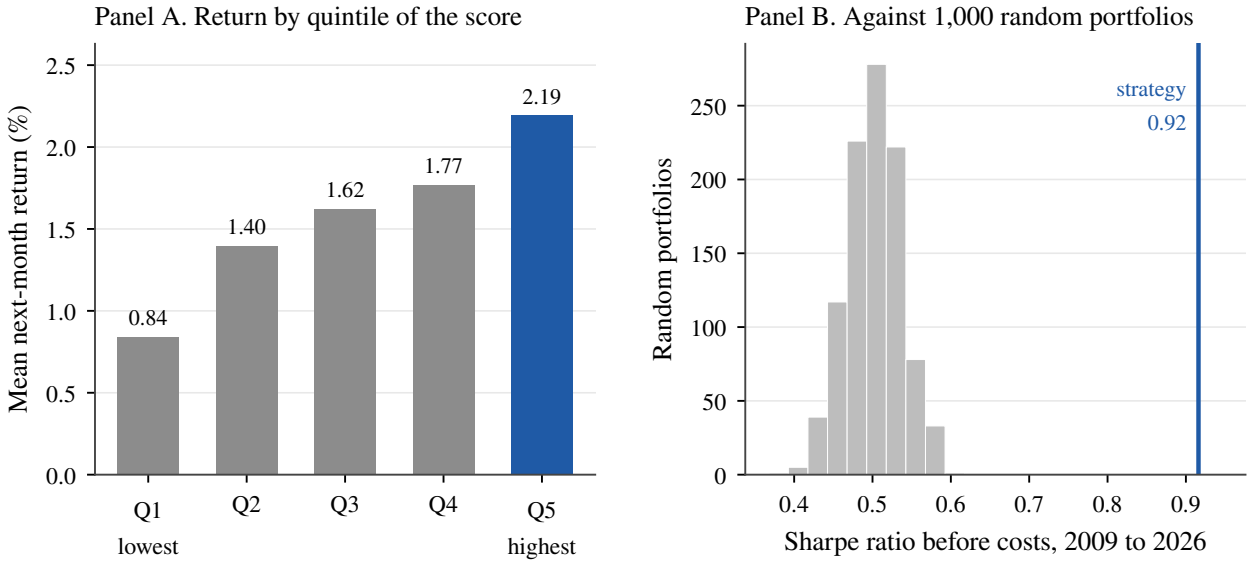


Figure 9. Panel A shows the mean next-month return of each quintile of the score, equal-weighted and before costs, March 2009 to June 2026; the strategy holds the top quintile. Panel B shows the Sharpe ratios of 1,000 random portfolios of the same size, before costs, and the strategy's.

4.7 What the strategy earns

Table 8 regresses the excess returns on the four factors of Agarwalla et al. (2013). The strategy's return over EW loads on momentum with a coefficient of 0.52 and leaves an alpha of +0.2% a year (t = 0.09). The edge over the benchmark is the momentum premium. The study does not find a new anomaly; it shows that a documented one survives Indian trading costs, a one-day execution lag and a correction for survivorship, in a form that a long-only investor can hold.

Table 8
Four-factor attribution on the survivorship-free universe

	Alpha, a year (t)	Market	Size	Value	Momentum (t)	R^2
Strategy	−1.1 (−0.47)	+1.04	+0.39	+0.28	+0.39 (8.5)	0.88
Equal-weight universe	+0.3 (0.25)	+1.12	+0.34	+0.21	−0.17 (−6.4)	0.97
Strategy minus EW	+0.2 (0.09)	−0.04	+0.05	+0.04	+0.52 (14.2)	0.70

Monthly returns from March 2009 to December 2025, regressed on the market, size, value and momentum factors of Agarwalla et al. (2013). The strategy and EW are in excess of the risk-free rate; the third row is the difference between them. Alphas are annualised; t-statistics use Newey-West standard errors.

NSE's own momentum index puts the result in proportion. The Nifty500 Momentum 50, which holds the 50 stocks with the strongest risk-adjusted momentum and weights them by momentum and size, returns 21.9% a year over the same period with a Sharpe ratio of 0.79 (Table 5). The strategy returns more, 23.8%, with the same risk-adjusted return, 0.80. It lagged the index in the development period and led it in the holdout. The index pays no trading costs, and its history before its launch in 2024 was computed by NSE with a method chosen in 2024. The fair summary is that the strategy captures the momentum premium about as well as a published momentum index, not better.

5. Robustness and weak points

Table 9 reports variants of the strategy, each changing one setting. None was used to choose the primary strategy.

Table 9
Variants of the strategy on the survivorship-free universe, net of costs

	Full sample			Holdout
Variant	CAGR	Sharpe	Over EW (t)	Over EW (t)
Primary: composite, top quintile	23.8	0.80	+5.2 (1.43)	+8.3 (2.39)
Rank buffer (hold until a name leaves the top 30%)	25.3	0.85	+6.4 (1.83)	+8.5 (2.56)
Fama-MacBeth forecast in place of the composite	21.9	0.73	+3.5 (1.04)	+8.0 (2.82)
Top decile	25.4	0.80	+6.9 (1.62)	+10.9 (2.48)
Top tercile	22.4	0.77	+3.8 (1.44)	+5.5 (2.05)
Momentum over 6 months	23.2	0.76	+4.7 (1.46)	+7.7 (2.43)
Momentum without the skipped month	23.8	0.81	+5.0 (1.39)	+8.1 (2.36)
Costs doubled	20.6	0.68	+2.5 (0.68)	+5.6 (1.61)
200 most traded stocks	18.8	0.63	+1.6 (0.46)	+5.9 (1.60)
1,000 most traded stocks	28.8	0.93	+7.5 (2.33)	+6.7 (2.22)
Distress exits at 0%	23.8	0.80	+5.1 (1.43)	+8.3 (2.38)
Distress exits at −100%	23.8	0.80	+5.2 (1.44)	+8.3 (2.39)

Each row changes one setting of the primary strategy. The rank buffer keeps a stock until it leaves the top 30% of scores. The Fama-MacBeth variant weights the four signals by the expanding mean of past cross-sectional regression slopes. The universe-size and exit-return rows rebuild the universe or the exit returns and rerun the strategy; they were specified before the replication was run. Return over EW is in percent a year with a Newey-West t-statistic.

The edge is outside the largest stocks. Within the most liquid third of the universe, the strategy's return over the same third's equal-weighted portfolio is +0.9% a year ($t = 0.25$), against +7.6% ($t = 2.16$) in the middle third and +6.8% ($t = 1.76$) in the least liquid third (Figure 10, Panel A). Restricted to the 200 most traded stocks, the return over EW falls to +1.6% ($t = 0.46$), and over a wider universe of 1,000 it rises to +7.5% ($t = 2.33$). This matches the finding of Hong et al. (2000) that momentum is stronger among smaller stocks.

Momentum crashes. The strategy trailed EW in 3 of 18 calendar years. The largest shortfalls came when the market turned sharply up after a fall. From March to December 2009 the strategy returned +83.2% while EW returned +173.2%, as the stocks that had fallen furthest in 2008 led the recovery, the pattern Daniel and Moskowitz (2016) document. In 2025 the strategy returned −11.0% against +1.9%. Scaling the long-short spread by its recent volatility, as Barroso and Santa-Clara (2015) propose, is the standard remedy; I report it as a variant of the spread and did not apply it to the long-only strategy.

Turnover and costs. One-way turnover is about 417% a year, so the strategy replaces its whole portfolio about four times a year, and costs take 2.6% a year. It still beats EW at twice the cost schedule (Table 9), but short-term capital gains tax, which applies to positions held for less than a year and would take a large share of the gains, is not modelled. The rank-buffer variant, which trades less, returns 25.3% with a Sharpe ratio of 0.85, but it was not the pre-specified primary strategy and its slightly better result is not evidence of anything.

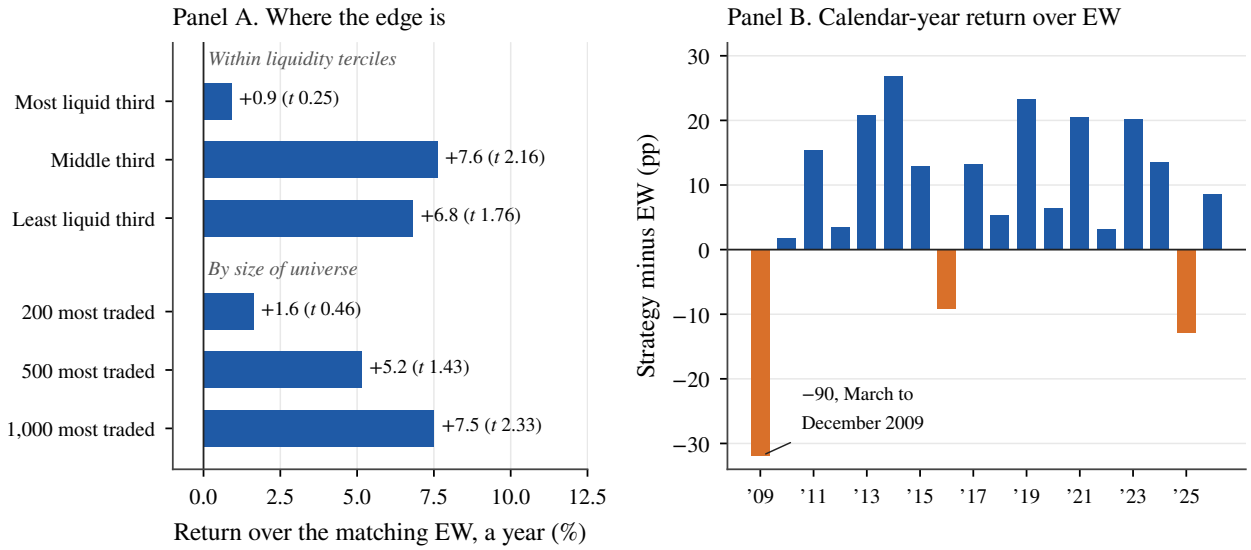


Figure 10. Where the edge is and when it fails. Panel A shows the return over the matching equal-weighted benchmark, a year, within each third of the universe by liquidity and for universes of three sizes, with Newey-West t-statistics. Panel B shows the strategy's calendar-year return minus EW's; 2009 runs from March and is cut at −32 points, and 2026 ends in June.

Exit returns barely matter. Setting every distress exit to 0% or to −100% changes the full-sample return over EW from +5.2% to +5.1% and +5.2%. Of the 688 stocks that stopped trading after March 2009, 53 were in the benchmark at the month-end before, 3 of them classed as distress, and 18 were held by the strategy, 0 as distress. A company on its way to failure loses liquidity long before it stops trading, and leaves the universe first.

The long-short spread. The return of the top quintile minus the bottom quintile is 13.1% a year with a Sharpe ratio of 0.39 and a maximum drawdown of −55.9%. The long side does the work; the spread is reported as a research measure, since it cannot be held in the Indian cash market.

Signal selection on the corrected data. Applied to the survivorship-free development data, the selection rule picks 7 signals, the four of the strategy and three more (Figure 11). The strategy keeps the four it was fixed with. Testing the larger set would be a new search on data whose holdout has already been seen.

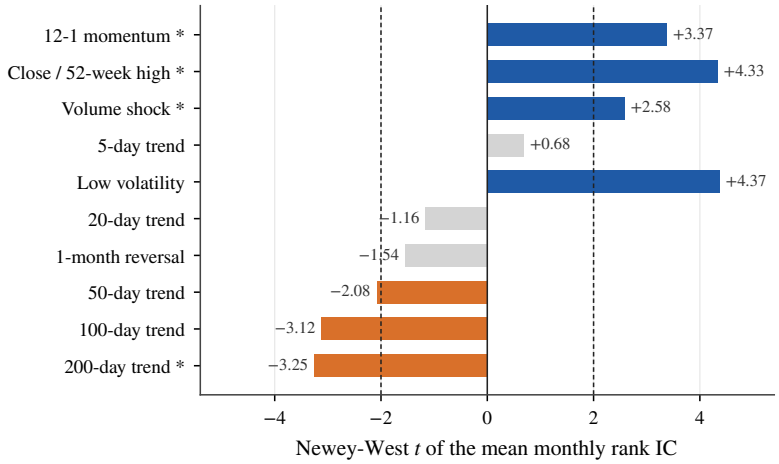


Figure 11. The selection rule of Section 3.2 applied to the survivorship-free development data. Signals beyond the dashed lines at ±2 would be selected, in blue with a positive sign and orange with a negative one; an asterisk marks the four signals of the strategy. Reported only: the strategy keeps the signals it was fixed with.

6. Discussion

The two parts of the study use the same kind of information, recent prices, and reach opposite answers. The difference lies in how the information is used. Part I asked whether a stock's recent prices forecast its own return tomorrow. Part II asked whether stocks with strong recent prices outperform stocks with weak ones over the next month. The first is a forecast of noise-dominated daily changes, judged one stock at a time; the second is a comparison across stocks at the horizon at which Jegadeesh and Titman (1993) found momentum, and it pools the evidence across hundreds of stocks and two hundred months. The null result of Part I is consistent with the weak form of the efficient market hypothesis (Fama, 1970) at the daily horizon. The result of Part II is consistent with a momentum premium that has persisted in India, as the positive momentum factor of Agarwalla et al. (2013) shows, and that carries crash risk, which is one explanation for why it has not been traded away.

What the evidence supports is a long-only strategy that captures that premium net of realistic costs, from mid-sized and less liquid stocks more than from the largest, with a return over its benchmark that is significant in the holdout and over the full sample by alpha, but not by the simple difference over the full sample. It does not support a claim of returns above 30% a year, which came from a universe of survivors.

7. Limitations

- *The holdout was seen before Part III.* The survivorship-free run used the strategy exactly as fixed in Part II, but its holdout years were no longer unseen, so it is a replication rather than a second out-of-sample test.
- *Prior knowledge.* The candidate signals were chosen from a literature that documents momentum, including Indian evidence that covers part of the holdout period. The holdout was never used to fit anything, but the choice of signal family was not blind to how such signals have performed.
- *Data rules.* The rules that clean and adjust prices were written while looking at the whole sample. They were checked against independent sources and fixed before any strategy return was computed, but they reflect judgements: the exit classes are partly inferred from each stock's own record, a handful of pre-2010 splits in then-illiquid stocks remain unadjusted, and about one dividend record in ten after 2010 gives no amount.
- *Universe definition.* The universe ranks stocks by traded value rather than by the market capitalisation that defines the Nifty 500, and the cost schedule of October 2024 is applied to the whole sample.
- *Costs not modelled.* Short-term capital gains tax and the price impact of large orders are not deducted, and both would matter most for the less liquid stocks where the edge is largest.

8. Conclusion

Four moving-average features in a linear model do not forecast the next day's return of 98 NSE stocks better than benchmarks that use no model. The same kind of information, when used to rank stocks against each other once a month, does carry a premium. Holding the top fifth of a survivorship-free universe of 400 stocks returned 23.8% a year from 2009 to 2026 net of costs, against 16.8% for an equal-weighted portfolio of the same stocks, and beat that benchmark by +8.3% a year in a holdout that was run once. The excess return is the momentum premium, about as large as that of NSE's momentum index and concentrated outside the largest stocks. A universe of today's survivors would have added about 8 points a year to every return in the study, which is the main lesson for anyone backtesting on Indian equities.

Natural extensions are a universe ranked by market capitalisation, tax and market-impact costs, signals that do not depend on price, such as earnings revisions and delivery volumes, and a forward test that records the strategy's holdings each month from July 2026.

Use of AI tools

While the research question, the ideas brought in and the interpretation of the results are my own, I used Claude Code, Anthropic's AI coding assistant, throughout this paper to learn more about computing, machine learning, and quantitative finance. In Part I, it mainly acted as a tutor and wrote parts of the code under my direction. In Parts II and III, Claude almost acted as a peer; it shared ideas, criticised mine and helped me find a solution to a complex problem I could not solve on my own. Artificial intelligence has redefined "learning by doing" to a whole new extent; I have learned more about computing and quantitative finance through the implementation of this project compared to what any theory or literature on this subject would have taught me. Technical terms and concepts in the code and in finance that I was not familiar with, I learned through the explanations Claude provided. The process of reviewing the code, reporting the results and documenting the project alongside Claude's explanations gave me a system where I could implement what I learned in a practical manner, which made this project more insightful. I checked Claude's explanations against the data before relying on them. I reviewed everything it produced and take full responsibility for the content.

Data and code availability

The project repository, https://github.com/arjunprashanth6129/price-prediction-linear-regression, holds the prices, code and results. Part I is in `scripts/ml/`, and Parts II and III are in `scripts/strategy/`, with the exchange files of Part III in `scripts/strategy/raw/nse/`. Running `python paper/make_assets.py` re-estimates every Part I model, checks the results against the committed output, and regenerates all numbers, tables and figures in this paper from the committed results of the strategy pipeline.

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A. Part I: largest and smallest edges

Table 10
Stocks with the largest and smallest edge over the always-up rule

Rank	Ticker	Band	Accuracy (%)	Up days (%)	Edge (pp)	R^2_{OS}	Binomial p
1	RPOWER	Small	53.9	44.4	+9.5	−0.072	0.16
2	GARFIBRES	Mid	51.4	42.5	+9.0	+0.014	0.63
3	HIMATSEIDE	Small	52.7	43.8	+8.9	−0.014	0.33
4	PAYTM	Large	54.2	45.3	+8.9	−0.084	0.23
5	CHOLAFIN	Large	53.3	46.1	+7.2	−0.005	0.24
6	GREENPANEL	Small	51.2	44.2	+7.0	+0.008	0.70
7	SUPRAJIT	Small	52.9	46.2	+6.6	+0.014	0.31
8	BALKRISIND	Large	50.3	43.9	+6.4	−0.002	0.96
			…				
91	KARURVYSYA	Small	47.6	50.4	−2.9	−0.054	0.39
92	HAVELLS	Large	45.5	48.4	−2.9	−0.011	0.11
93	DRREDDY	Large	47.8	51.9	−4.0	−0.006	0.45
94	NAVINFLUOR	Mid	48.6	52.6	−4.0	−0.059	0.63
95	FEDERALBNK	Large	47.6	52.2	−4.6	−0.025	0.39
96	ALKEM	Large	47.1	51.7	−4.6	−0.012	0.31
97	BAJAJCON	Small	46.4	51.3	−4.9	−0.068	0.20
98	LAURUSLABS	Large	46.7	55.6	−8.9	−0.130	0.24

Stocks ranked by edge, the model's directional accuracy minus the share of up days in the evaluation window, in percentage points. The binomial p-value tests accuracy against 50% and is not adjusted for multiple testing. Full results for all 98 stocks are in `results/per-stock.csv` in the repository.